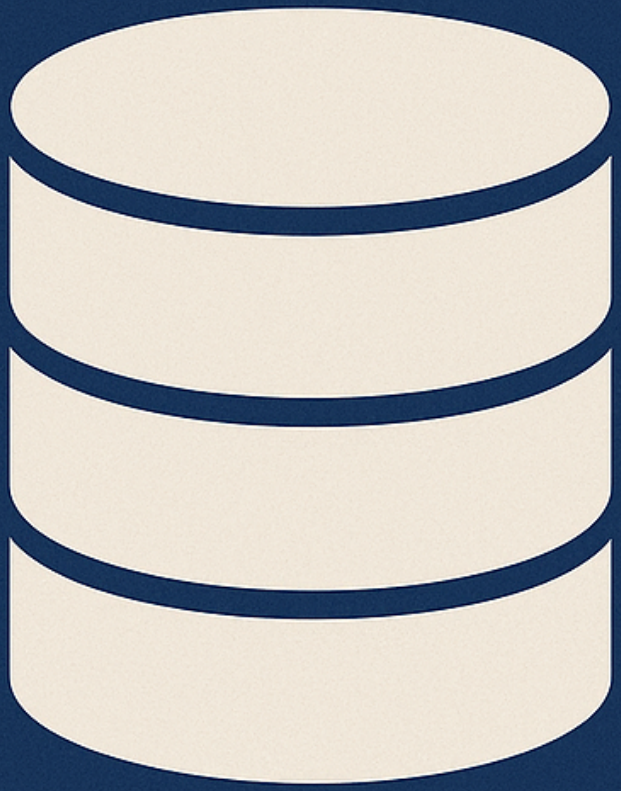
SQL

Class starts
@ 9:10pm





1) What is Adaptive Batch?

→ 2 classes / week & to be in a AB a certain % PSP
↳ problem solving percentage

"SQL" ⟶ "Product Analytics"

Data Science
(Data Analyst / Business Analyst
/ product Analyst / Data Scientist)

WhatsApp Group → "Break" / "Next Class"

Lecture 1: Introduction to database and Bigquery Setup

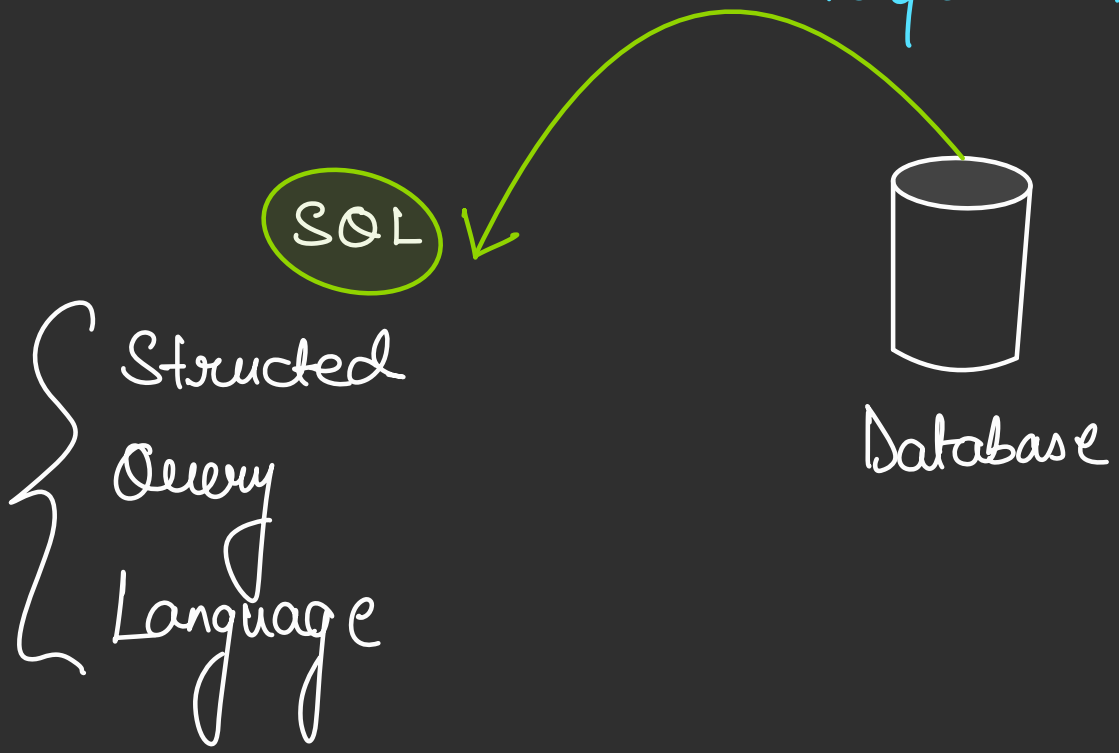
#Agenda

1) Module overview
2) What is Database?
3) DBMS, Database Vs Excel
4) DB Schema Design
5) Concepts of Keys
6) Data Warehouses
7) Setting up big query.

Importance

Company & Role	Responsibilities	Skills Required
Agoda, Data Analyst	You'll be heavily involved in growing our Flights business Experiment with various pricing strategies on Agoda and Metasearch to grow the business Analyze massive amounts of data generated in experiments, build dashboards, and develop predictive models	Good numerical reasoning skills Strong problem-solving skills Proficiency in Excel and **SQL** Exposure to one or more data analysis packages or databases, e.g., Python, R, **SQL**
IBM, Data Analyst	At the strategic level, you will help design the processes & infrastructure that underlie IBM's Marketing goals. and execute elements of our strategic data plan. At the tactical level, you will help our user communities to utilize data efficiently, maximizing efficiency & effectiveness. You will also interface with service providers at the infrastructure, development and operations levels on a daily basis and IBM Product & Service support teams as required.	**SQL** and Analysis tools such as dBeaver, Python or other Query/Insight tools accessing DB2, Big Data environments Experience with BI Tools such as Cognos, Tableau, etc. would be beneficial
Target, Data Analyst	As a Data Analyst in Finance Capabilities partnering with the Global Supply Chain Logistics (GSCL) Finance team, you will work collaboratively with a global team to	Strong academic performance in a quantitative field **Intermediate SQL** experience writing complex queries

Helps to extract data from DB



#Module Overview {2 Months long}

1. **Introduction**

   Database Fundamentals, Data Types, Keys, SQL vs Excel etc.

2. **Data Extraction**

   Extracting Data: SELECT, ORDER BY, and LIMIT
   Filtering Data: WHERE, AND, OR, NOT, IN, BETWEEN, LIKE, and wildcards
   Handling Logic : Subqueries, CASE statements, IF function

3. **Data Aggregation**

   Grouping and Analyzing Data: MIN, MAX, SUM, AVG, and COUNT functions. HAVING clause to filter aggregated data.

4. **Combining Data**

   Joins Types: INNER, LEFT, RIGHT, FULL, and UNION to combine data from multiple tables.
   Conquer Self-joins, Cross joins, and Non-equi joins for advanced scenarios.

5. **Advanced Data Analysis**

   Window functions: ROW_NUMBER, RANK, DENSE_RANK, and LAG/LEAD, Nth Value to analyse trends and compare data within groups. Learn to format, parse, and manipulate date and time values in SQL.

6. **Advanced Constructs**

   Create and leverage CTEs for complex queries. Understand Views and how they differ from CTEs.
   SQL Query Optimization techniques for efficient data retrieval

7. **MySQL Setup & Commands**

   Learn to set up MySQL Workbench and understand different types of SQL commands. Explore DDL & DML commands.

8. **Beyond Basics**

   Delve into advanced concepts such as partitioning, indexes, the EXPLAIN command, user-defined functions, stored procedures, correlated subqueries, and the EXISTS clause .. e.t.c.

9. **Target Case study**

   You will be working on the real time dataset to extract valuable insights and provide actionable recommendations.

10. **Module Test & Module Re-test**

DBMS and RDBMS

Relational DBMS

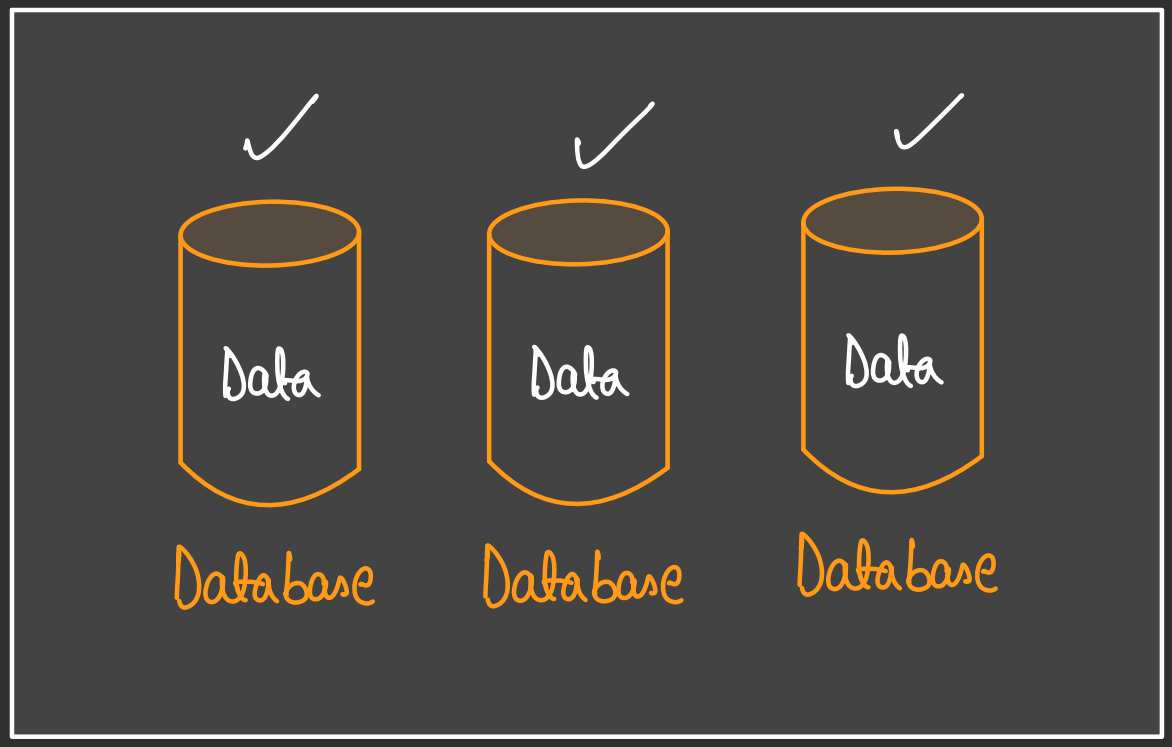
Database Management System:

Amazon Farmers Market (AFM) is a beloved community market that focuses on providing fresh, local produce directly from farmers to consumers.

Known for its commitment to quality and sustainability, AFM has become a vital hub for the community, offering a wide variety of fruits, vegetables, and artisanal products.

Despite its popularity, AFM faces increasing competition from large grocery chains that are expanding their local produce offerings.

To maintain its customer base and thrive in this competitive environment, AFM aims to leverage data analytics for a deeper understanding of customer behaviour, vendor performance, market operations, and inventory management.

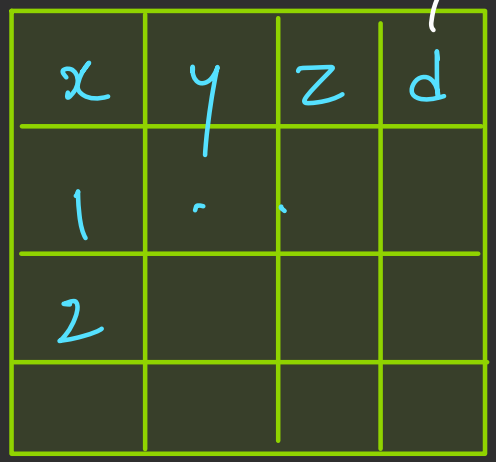


Management System

Database

(DBMS) → Collection of databases

Microsoft Excel



colourns

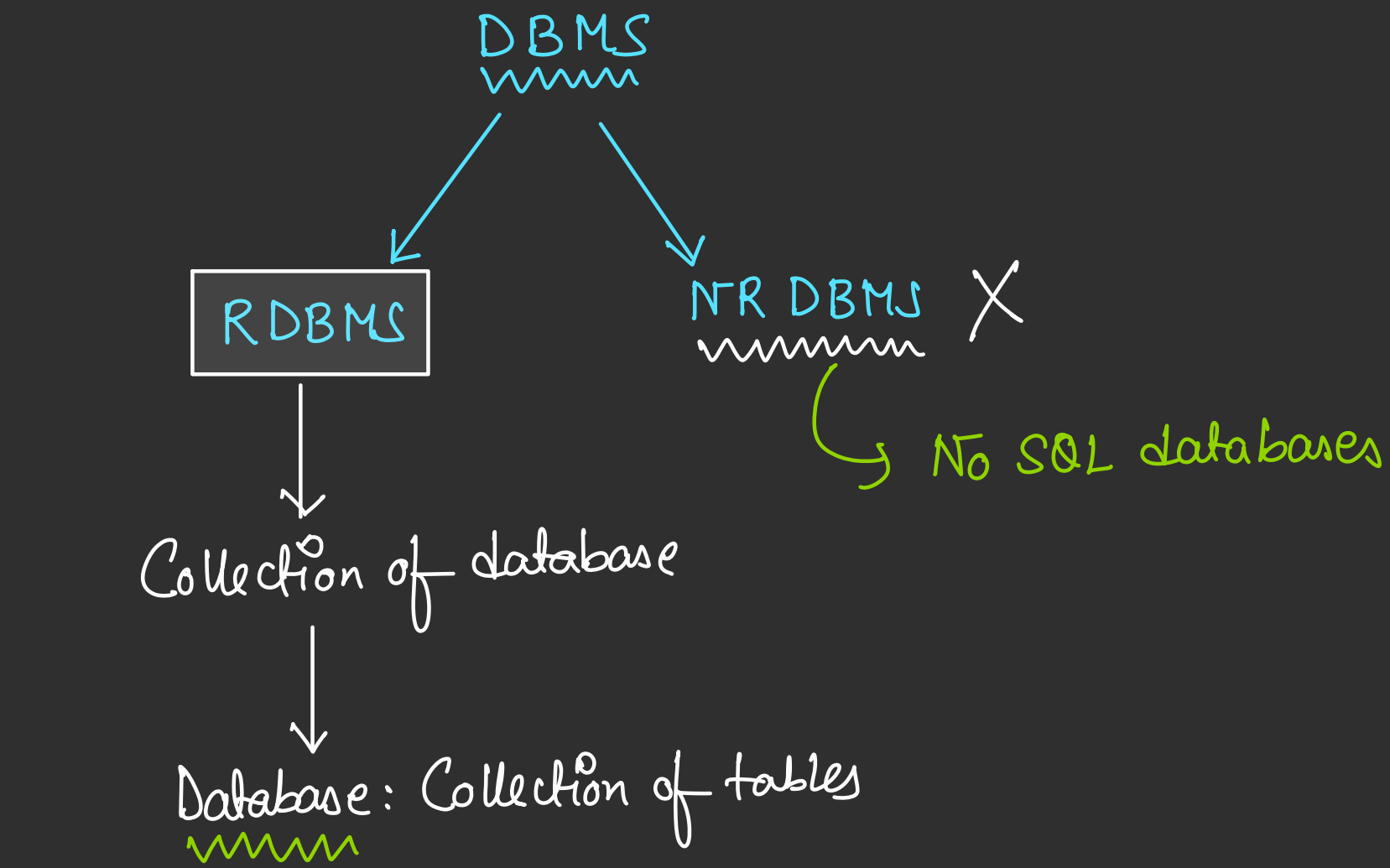
Rows

table

→ tabular Data

Relational data

RDBMS


DBMS
RDBMS
NR DBMS
X
No SQL databases
Collection of database
Database: Collection of tables

SQL

Most popular RDBMS are –

a) MySQL ✓
b) Oracledb
c) SQLite
d) BigQuery ✓
e) PostgreSQL

Excel Vs Database

1) Scalability → Excel only allows 1M records

2) Performance → Large data extraction / ingestion needs to be fast
   (ingestion = data input)

3) Data integrity → Database can control acess

4) Security → Security is high Database ↑

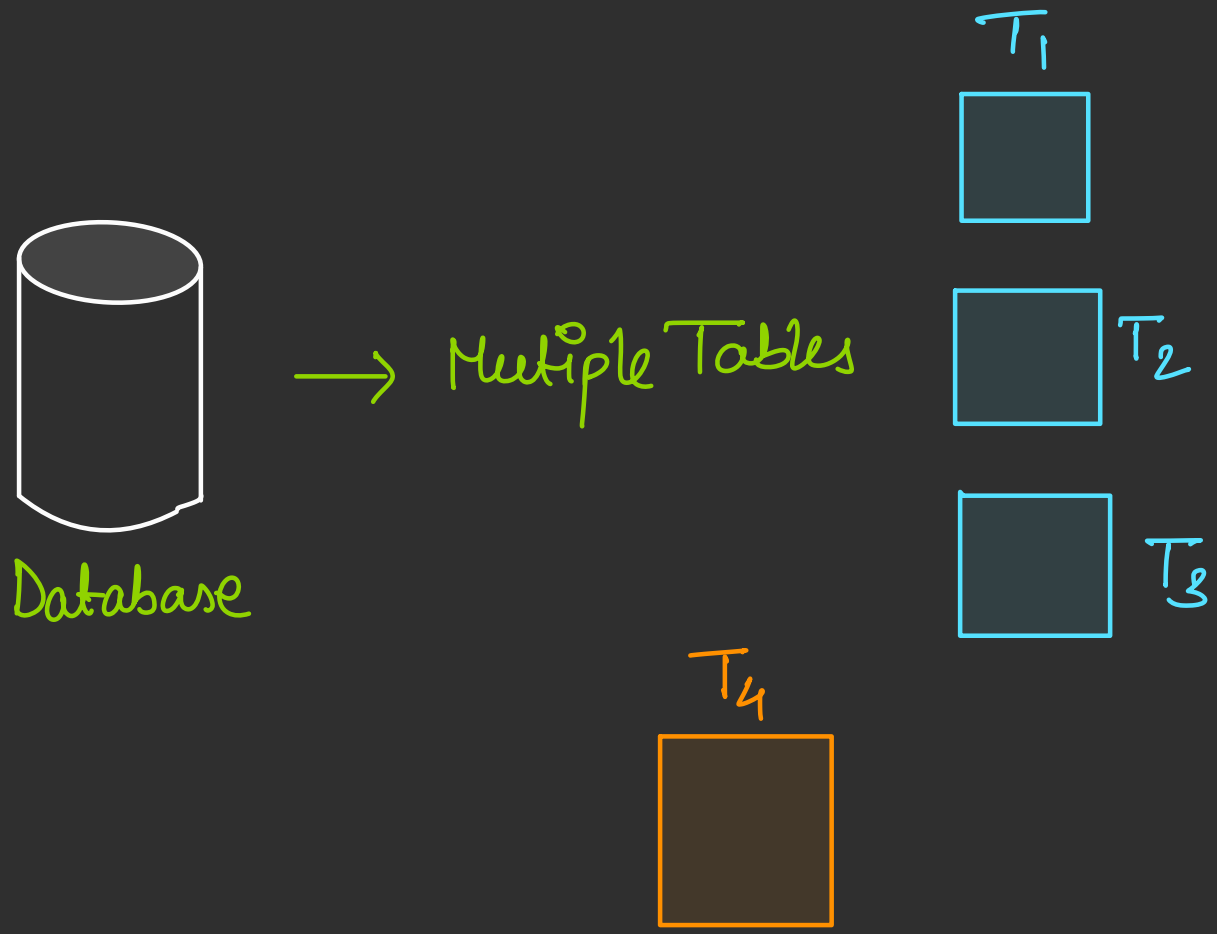
5) Concurrent Acess → Multi user access

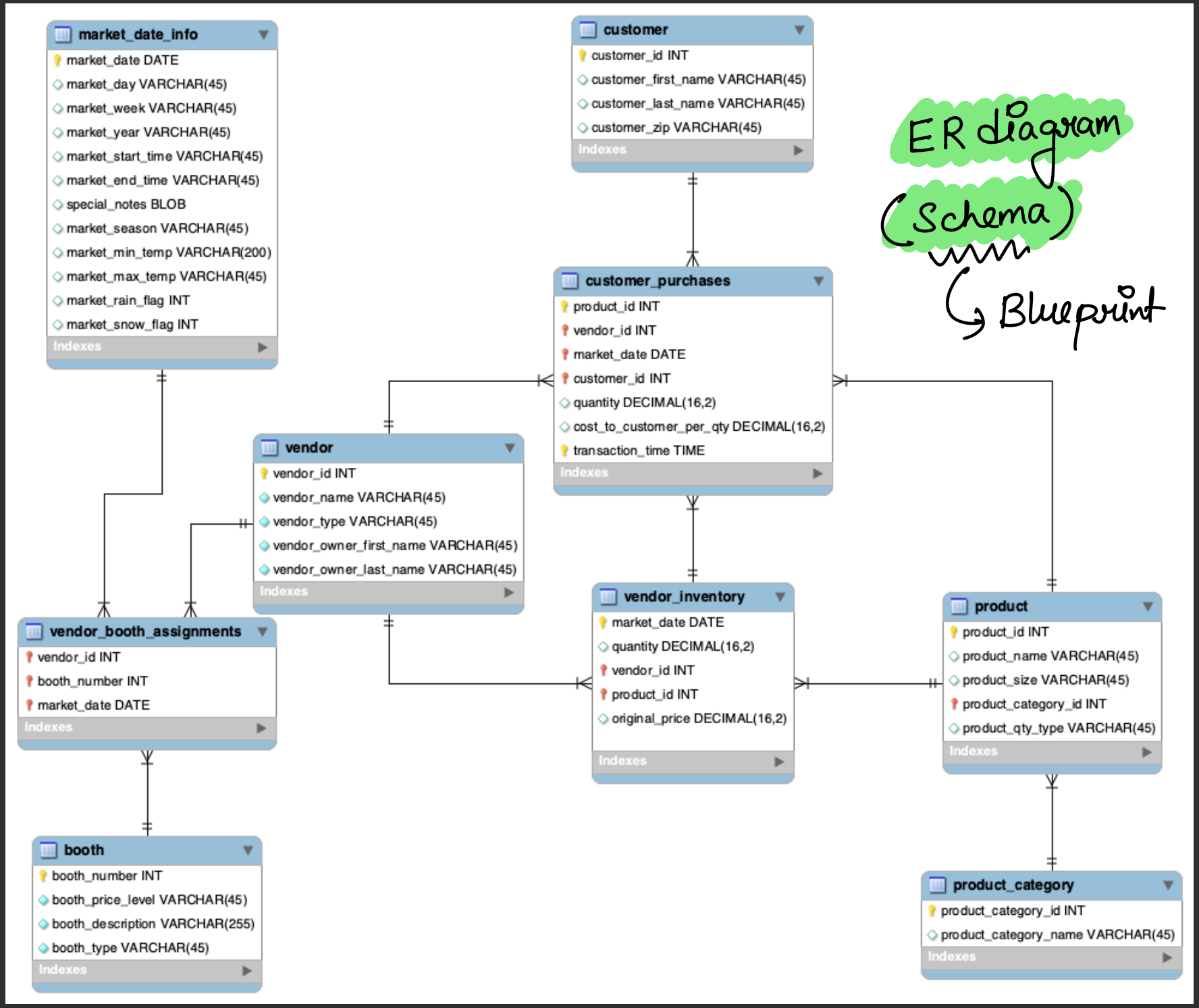
Amazon Farmers Market (AFM), a beloved community market, faces a growing threat from large grocery chains that are expanding their local produce offerings.

It’s your first day as a Data Analyst Intern at Amazon Fresh.

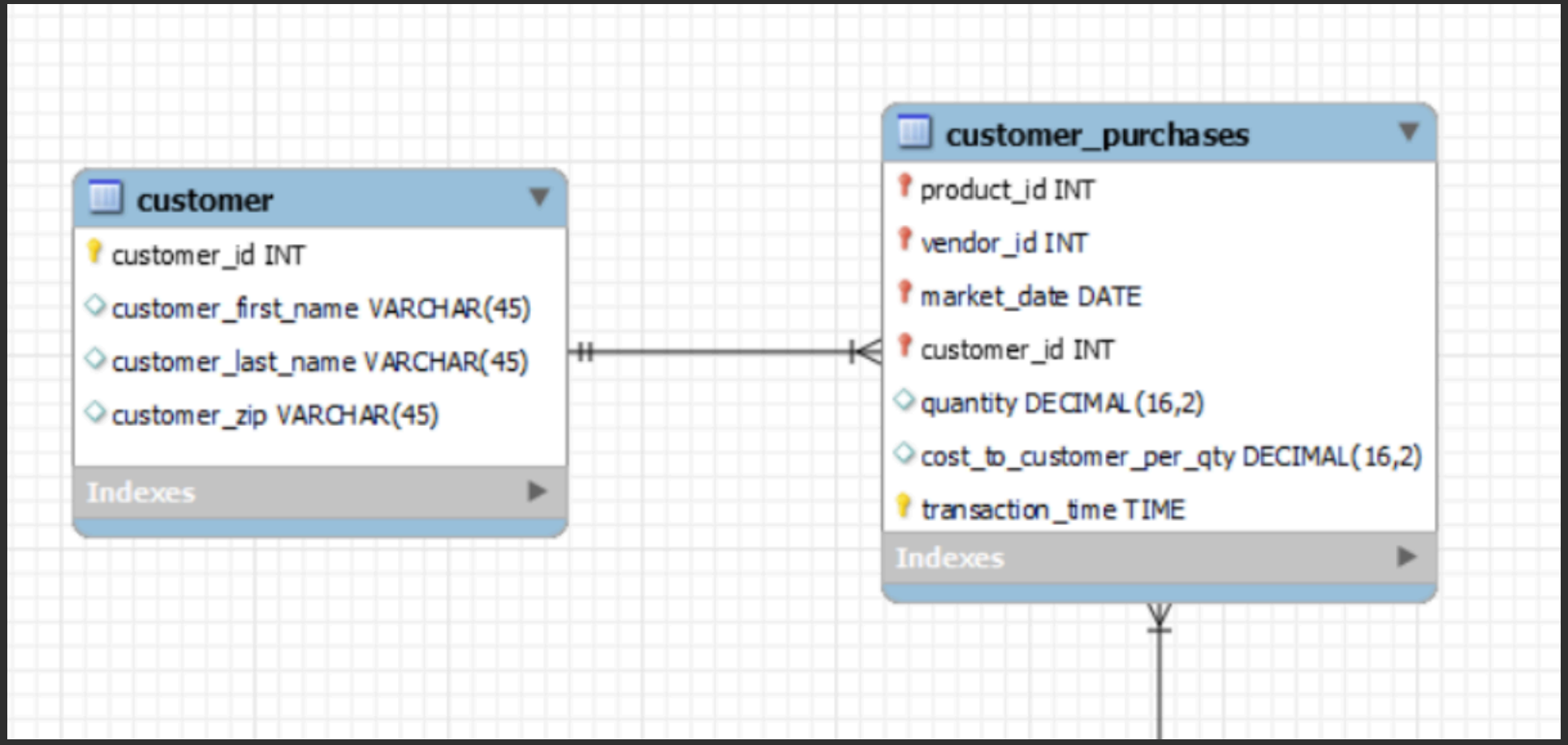
Amazon Fresh has recently decided to open big stores where farmers can directly sell their produce from their stores.

All the data is stored in a Database.


T1
Database
Multiple Tables
T2
T3
T4


market_date_info
market_date DATE
market_day VARCHAR(45)
market_week VARCHAR(45)
market_year VARCHAR(45)
market_start_time VARCHAR(45)
market_end_time VARCHAR(45)
special_notes BLOB
market_season VARCHAR(45)
market_min_temp VARCHAR(200)
market_max_temp VARCHAR(45)
market_rain_flag INT
market_snow_flag INT
Indexes
customer
customer_id INT
customer_first_name VARCHAR(45)
customer_last_name VARCHAR(45)
customer_zip VARCHAR(45)
Indexes
ER diagram
(Schema)
Blueprint
customer_purchases
product_id INT
vendor_id INT
market_date DATE
customer_id INT
quantity DECIMAL(16,2)
cost_to_customer_per_qty DECIMAL(16,2)
transaction_time TIME
Indexes
vendor
vendor_id INT
vendor_name VARCHAR(45)
vendor_type VARCHAR(45)
vendor_owner_first_name VARCHAR(45)
vendor_owner_last_name VARCHAR(45)
Indexes
vendor_booth_assignments
vendor_id INT
booth_number INT
market_date DATE
Indexes
vendor_inventory
market_date DATE
quantity DECIMAL(16,2)
vendor_id INT
product_id INT
original_price DECIMAL(16,2)
Indexes
product
product_id INT
product_name VARCHAR(45)
product_size VARCHAR(45)
product_category_id INT
product_qty_type VARCHAR(45)
Indexes
booth
booth_number INT
booth_price_level VARCHAR(45)
booth_description VARCHAR(255)
booth_type VARCHAR(45)
Indexes
product_category
product_category_id INT
product_category_name VARCHAR(45)
Indexes

ER Diagram (Entity-Relationship) {ERD}



Data Types

Tables ⟶ row & column

Email, Age, Name, Joining date, Salary

(text) (Num) (text) (date) (Decimal) → Normal

(string) (integer) (date) (float) → program ways

13

13.62
13.0

primary key

customer_id	name	phone	age
1	Akon	9876723452	17
2	Akon	9991165674	19
3	Bkon	7898756543	18
4	Ckon	8987867898	19
5	Dkon	9990080080	//////

"Only 1 primary key per table".

Null value
(Absence of data)

Name = ? X

phone Number = X

Age = X

customer_id

Unique & Not Null

primary key

Quiz

customer_id	aadhaar_number
1	4738 2956 7103
2	9123 7456 8521
3	1357 8964 2053
4	8765 4321 0987
5	

Null Value

Hence AN can't be used for primary key?

primary key = Unique + Not Null

PK

Foreign key = Can have duplicates + Can have Null valu

order_id	item	quantity	cust_id
1	Fruits	2.5	1
2	Veggies	3	1
3	Meat	1	3
4	Veggies	5	5
5	Dairy	2	2

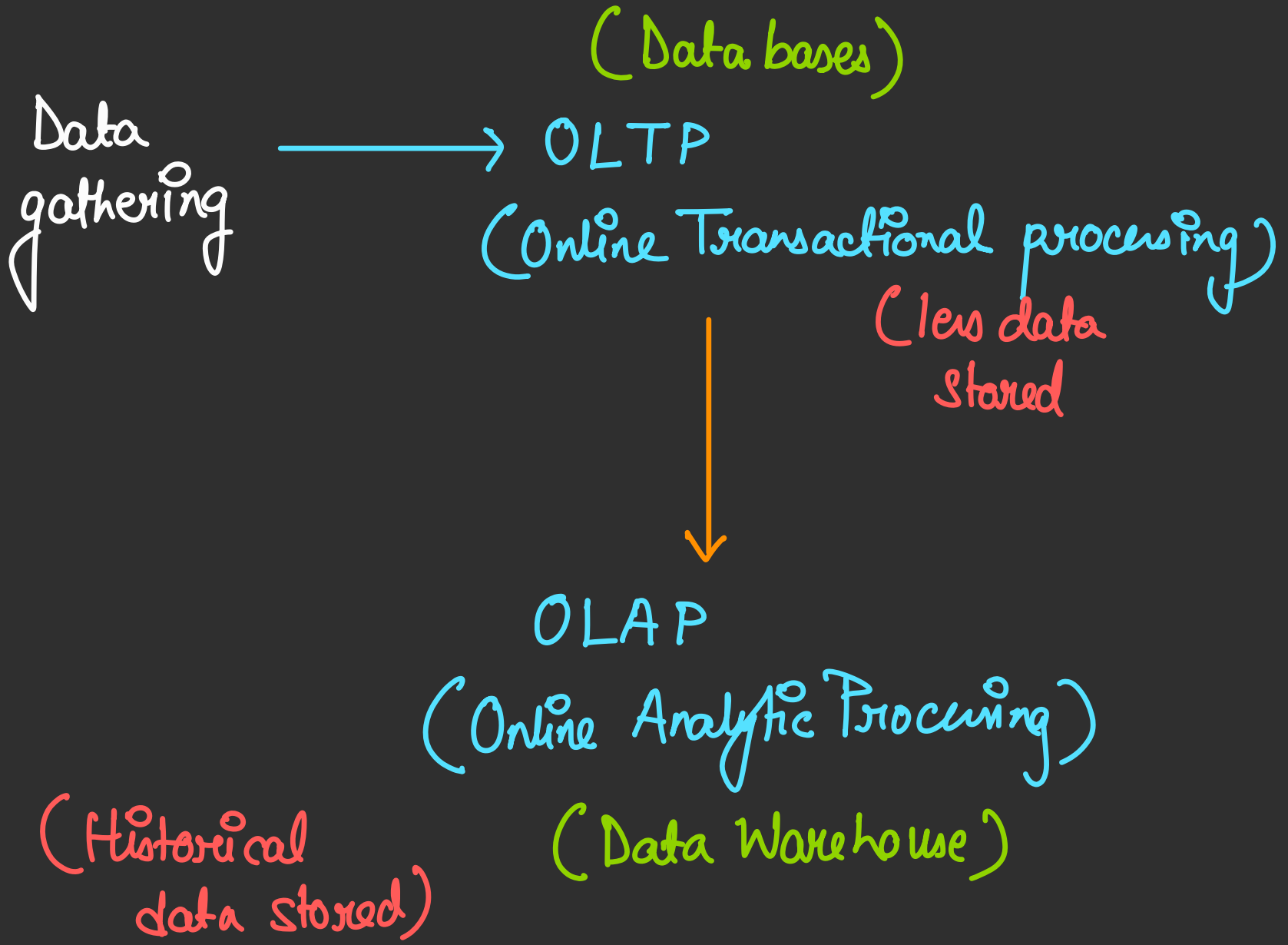
orders

PK

customer_id	name	phone	age
1	Akon	9876723452	17
2	Akon	9991165674	19
3	Bkon	7898756543	18
4	Ckon	8987867898	19
5	Dkon	9990080080	

customer

Data Base Vs Data Warehouse



id	Name
1	Ani
2	Ani

PK → "Name + Surname + Pho"

Logger

Machine_id	State	Date_updated
101	Online	31-10-25
102	offline	31-10-25
101	Offline	4-11-25

{ agg - groupby
o
vons -

⇒

101	Max date
102	Max date

↓

state for each mid

get the lastest status of each machine

101	Max date	Min date	⇒ Max - Min

DATEDIFF

Github

↳ Code storage platform

Repository (folders)